

Benchmark

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Invited: Benchmark: A Web-Based System for Tracking Experimental Results

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Abstract

Benchmarks have been a cornerstone of research in integrated

Physical Design (ISPD '26), March 15–18, 2026, Bonn, Germany. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3764386.3790088>

Background

More than twenty-five years ago, I started working on circuit placement (after having spent entirely too much time working on global routing).

To have an idea of what sort of results I'd need to get a paper published, I started gathering results from recent papers into an Excel spreadsheet. And nothing made sense.

ISPD 2001 "Reporting" paper

	[13] Gordian	[9] PRC	[22] HALO	[14] POPINS	[19] TW7.0	[10] QUAD	[18] NRG
Fract				45225		337	25602
Struct	558+362					3780	287631
Primary1	841+553	999.5	1.439	1128245	0.83	8972	894545
Primary2	4761+3153	3665.6	6.73	4128324	3.53	36824	3412195
Biomed	5232+3123			4128324	3.22	23765	
Industry1							
Industry2					13.30	332318	
Industry3					41.53	938682	
Avqsmall				23848188	5.08	62890	
Avqlarge				28323022	5.65	65906	
Golem3					88.98		
Units	X1000	X1000	Meter	Micron	Meter	X100	Micron
Spacing	Routed				Routed	None	

	[5] FD98	[7] ARP	[21] Dragon	[24] SPADE	[11] Mongrel	[23] Feng Shui	[12] iTools
Fract		0.034		0.024		0.032	
Struct	0.338	0.34		0.291	0.266	0.380	0.272
Primary1	0.87	0.79		0.74	0.83	1.018	0.799
Primary2	3.72	3.61		3.13	2.94	3.684	3.37
Biomed	1.78	1.83		1.43		1.689	2.90
Industry1		1.50				1.606	
Industry2	14.6		12.88	11.90	11.89	15.408	11.4
Industry3	45.1	48.12	42.33	35.37	34.53	44.729	39.6
Avqsmall	4.91	6.06	5.17	5.08	4.4	5.960	4.48
Avqlarge	5.38	6.54	5.25	6.16	4.87	6.301	4.78
Golem3			77.56	19.84		21.882	79.9
Units	Meter	Meter	Meter	Meter	Meter	Meter	Meter
Spacing	Row		Row	None	None	Row	Routed

<http://dx.doi.org/10.1145/369691.369727>

What Was Going On?

- Some placements have routing channels, others don't. Huge changes in HPWL for this.
- Different numbers of rows in placements
- Pad locations often moved substantially
- File format differences (YAL, VPNR, TW) could result in a 4X change in reported results

I emailed a dozen different groups to ask how they computed their results. Everyone replied with a variation of "we use the standard methods."

Surprise. There were no "standard methods."

How Did This Happen?

This was the late 1990's. The web had been around for only a few years. Circuit benchmarks were often distributed on magnetic tapes. If you wanted to submit a paper to DAC, you had to snail mail physical printed copies. Or drive to the LAX airport, because the FedEx office stayed open late.

Many people knew that the situation wasn't ideal, and that there was miscommunication. I said **the quiet part, out loud**, and found that while it was a little uncomfortable, there was broad agreement that we could and should do things better.

ISPD 2001

5. SUMMARY AND CONCLUSION

Circuit placement is a central problem in VLSI design automation. While fabrication technology has advanced at a breakneck pace, design tools have had a difficult time keeping up.

In this paper, we have shown that there is wide variation in the interpretation of placement benchmarks. This has resulted in vast differences in reported results, with many instances of “apples” being inadvertently compared to “oranges.” The placement problem is extremely complex, and as we have shown, many research groups have made assumptions that, while reasonable, differ substantially from the assumptions of others.

Improvement to design automation research is of profound importance; it is difficult to imagine how this will occur if we cannot accurately evaluate or compare approaches. One of the purposes of publishing research is to enable others to understand the merit of an approach, and to make subsequent comparisons. Without a common interpretation of the available benchmarks, the entire research process is undermined.

ISPD 2001

5.1 Distribution of Placement Results

We would encourage research groups working on placement to make their placement output available (perhaps through the web). This is common practice in many other academic areas, and we expect that this would reduce ambiguity in reported results substantially. Distribution of results requires very little effort.

As part of the GSRC Bookshelf[8] effort, placement results from a number of tools have been made available. In addition, both source code and executable versions of some placement tools are also available, allowing independent verification of results.

We have refrained from presentation of “definitive” placement results in this paper, preferring instead to direct researchers towards a Web-based table of current results. Placement results are large, preventing unambiguous description in the limited space available for a technical paper. Additionally, placement tools are being continually improved, and results that could be reported here would likely become outdated quickly.

What Happened Next?

ISPD contests starting in 2005. Benchmark suites were created each year, and many groups submitted tools. The tools were tested by the independent group, with all results being carefully checked.

Rapid progress, new ideas, experimental results that were the gold standard. Good times.

20th Anniversary of Making Everyone Angry

Session 4: Driving Research in Placement: A Retrospective

ISPD '21, March 22–24, 2021, Virtual Event, USA

Still Benchmarking After All These Years

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25th Anniversary? Benchmarker Web Site

The tracking of research results that I had hoped for didn't really materialize. In 2001, the web was unbelievably annoying to work with.

Times have changed.

<https://cs.binghamton.edu/~pmadden/benchmarker>

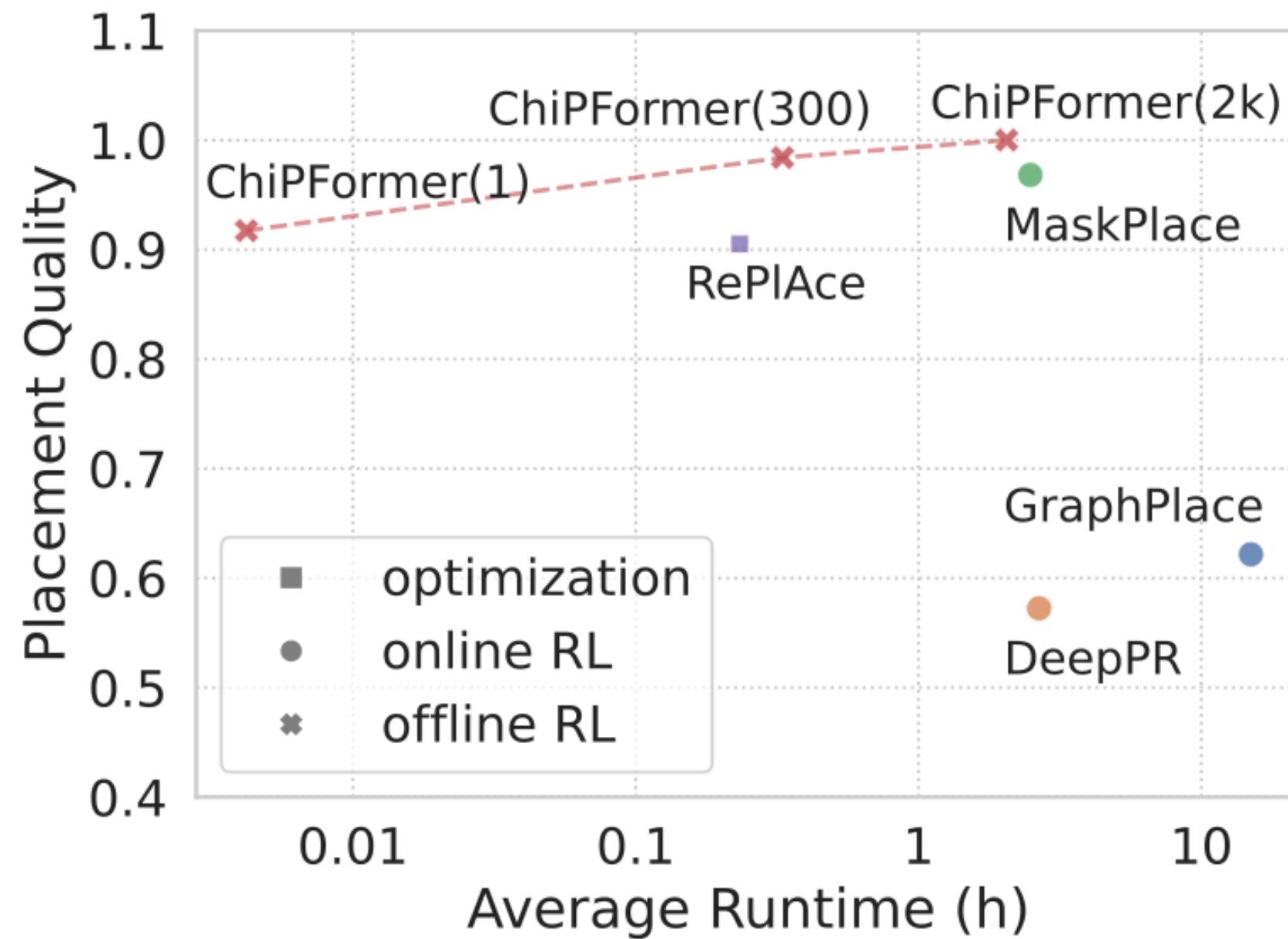
Now, the tricky part of the talk....

Artificial Intelligence is taking over everything

- Need speech-to-text for this talk? AI!
- Need to polish the text of a paper? AI!
- Need a bunch of citations to related work? AI!
- Need a homework assignment completed? AI!
- Need a better clothes washing machine? AI!
- Need a better electric toothbrush? AI!

....

- **Need a better circuit placement?**



ChiPFormer ICML 2023, <https://icml.cc/virtual/2023/poster/25027>

Table 8: **Comparisons of HPWL ($\times 10^7$) for mixed-size placement in *ICCAD04* benchmark.** HPWL is the smaller the better. SA (simulated annealing) method is implemented as in [Mirhoseini et al. \(2021\)](#). ChiPFormer(workflow) can achieve the best placement quality.

circuit	SA	RePlAce	GraphPlace	MaskPlace	ChiPFormer(workflow)
ibm01	25.85	<u>22.82</u>	31.71	24.18	16.70 (-26.82%)
ibm02	54.87	<u>47.59</u>	55.11	<u>47.45</u>	37.87 (-20.19%)
ibm03	80.68	<u>64.36</u>	80.00	71.37	57.63 (-10.46%)
ibm04	83.32	<u>72.61</u>	86.86	78.76	65.27 (-10.11%)
ibm06	69.09	<u>58.07</u>	63.48	<u>55.70</u>	52.57 (-5.62%)
ibm07	111.03	<u>98.57</u>	117.70	<u>95.27</u>	86.20 (-9.52%)
ibm08	131.07	<u>114.67</u>	134.77	120.64	102.26 (-10.82%)
ibm09	135.45	<u>120.01</u>	148.74	122.91	105.61 (-12.00%)
ibm10	423.14	<u>274.29</u>	440.78	367.55	230.39 (-16.00%)
ibm11	210.12	<u>169.98</u>	218.73	202.23	160.60 (-5.52%)
ibm12	410.05	<u>306.33</u>	438.57	397.25	273.14 (-10.83%)
ibm13	259.89	<u>220.14</u>	278.92	246.49	197.20 (-10.42%)
ibm14	405.80	<u>341.80</u>	455.32	<u>302.67</u>	301.28 (-0.46%)
ibm15	510.06	<u>451.36</u>	520.06	457.86	429.71 (-4.80%)
ibm16	614.54	<u>516.05</u>	642.08	584.67	463.32 (-10.22%)
ibm17	720.40	<u>635.93</u>	814.37	643.75	569.13 (-10.50%)
ibm18	442.00	<u>399.43</u>	450.67	<u>398.83</u>	370.36 (-7.14%)

ChiPFormer ICML 2023, <https://icml.cc/virtual/2023/poster/25027>

Chip Placement with Diffusion Models

Table 6. Comparison of HPWL (10^6) averaged over 5 seeds, using various techniques for mixed-size placement, on the IBM benchmark.

Circuit	MaskPlace + DP	WireMask-BBO + DP	ChiPFormer + DP	DREAMPlace	Diffusion (Ours)
ibm01	3.33	2.84	3.35	2.23	2.09
ibm02	7.30	6.87	6.24	5.79	4.43
ibm03	10.1	9.81	10.9	10.4	7.30
ibm04	10.4	9.65	10.1	9.13	8.00
ibm05	7.67	7.67	7.67	7.60	7.79
ibm06	7.62	8.41	7.76	6.15	8.31
ibm07	13.3	13.0	13.4	11.1	9.60
ibm08	15.5	15.9	15.7	12.3	13.3
ibm09	16.2	15.4	16.9	12.8	12.6
ibm10	46.8	45.2	45.4	44.8	30.2
ibm11	23.5	24.6	23.6	16.6	17.3
ibm12	46.1	Failed	48.8	31.0	34.0
ibm13	28.2	28.0	28.4	23.2	23.0
ibm14	45.4	48.2	46.5	31.3	34.5
ibm15	53.4	Failed	55.8	51.3	45.0
ibm16	65.9	63.2	67.3	53.0	52.7
ibm17	72.9	69.7	71.4	57.9	60.4
ibm18	42.2	41.6	41.1	37.6	38.6
Average	28.7	27.0	28.9	23.6	22.7

Diffusion, ICML 2025, <https://arxiv.org/pdf/2407.12282>

Which is it?

- ChiPFormer is heavily cited
 - ICML 2023 reports ChiPFormer as 1.67, while ICML 2025 reports ChiPFormer as 3.35. The established result for RePlAce is about 2.24.
 - Either it's **27% better** than RePlAce, or **54% worse**
 - *Kind of a big gap.*
- ChiPFormer placement results are not available.
- And there's one more gotcha...

Table 8: **Comparisons of HPWL ($\times 10^7$) for mixed-size placement in *ICCAD04* benchmark.** HPWL is the smaller the better. SA (simulated annealing) method is implemented as in [Mirhoseini et al. \(2021\)](#). ChiPFormer(workflow) can achieve the best placement quality.

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Note the exponent. It should be 10e5, not 10e7, if the RePlAce result is to make any sense. The authors confirmed that the exponent is a typo.

Anyone who reads the ChiPFormer paper, and then implements a placer... If the result is less than 100x worse than the results in the table, they might mistake it as a massive advance.

The ChiPFormer paper has been cited 90+ times in the past couple of years. To the best of my knowledge, no one else has noticed that the results could be off by 100x.

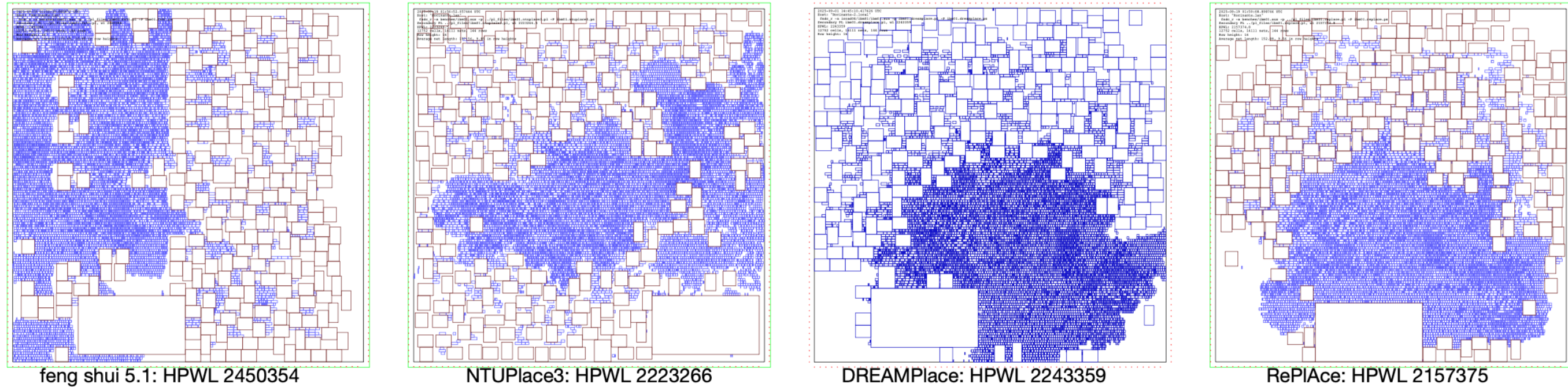
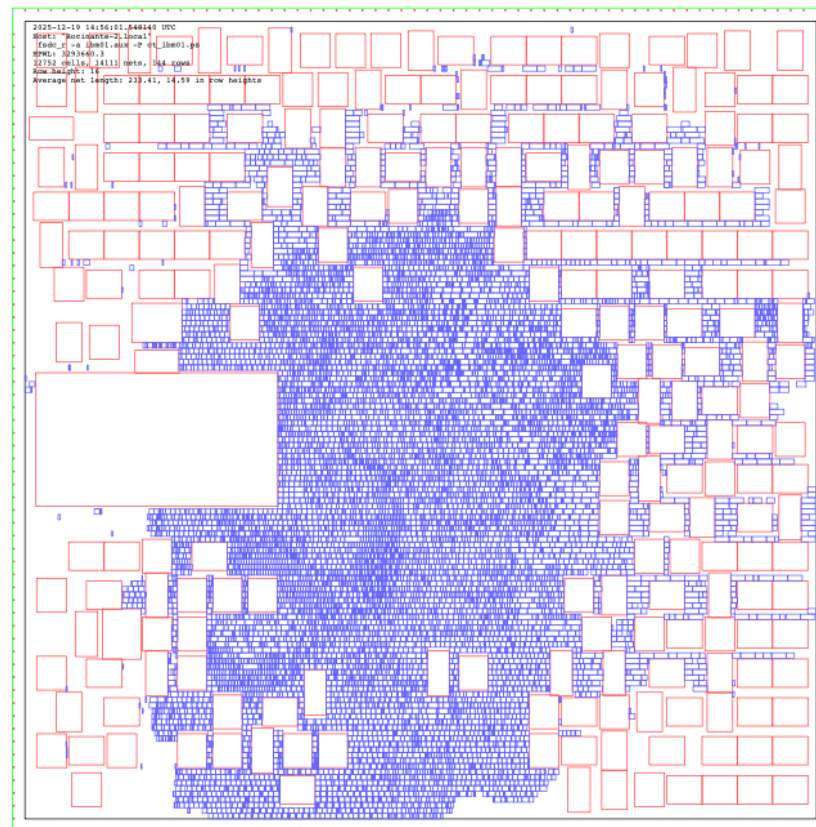
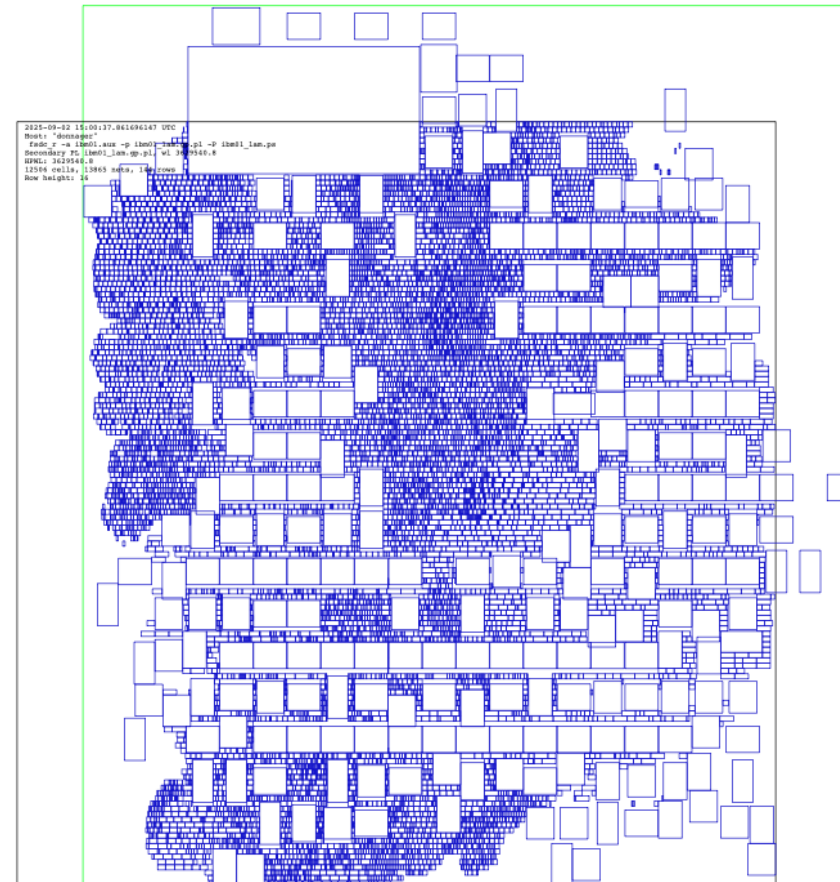


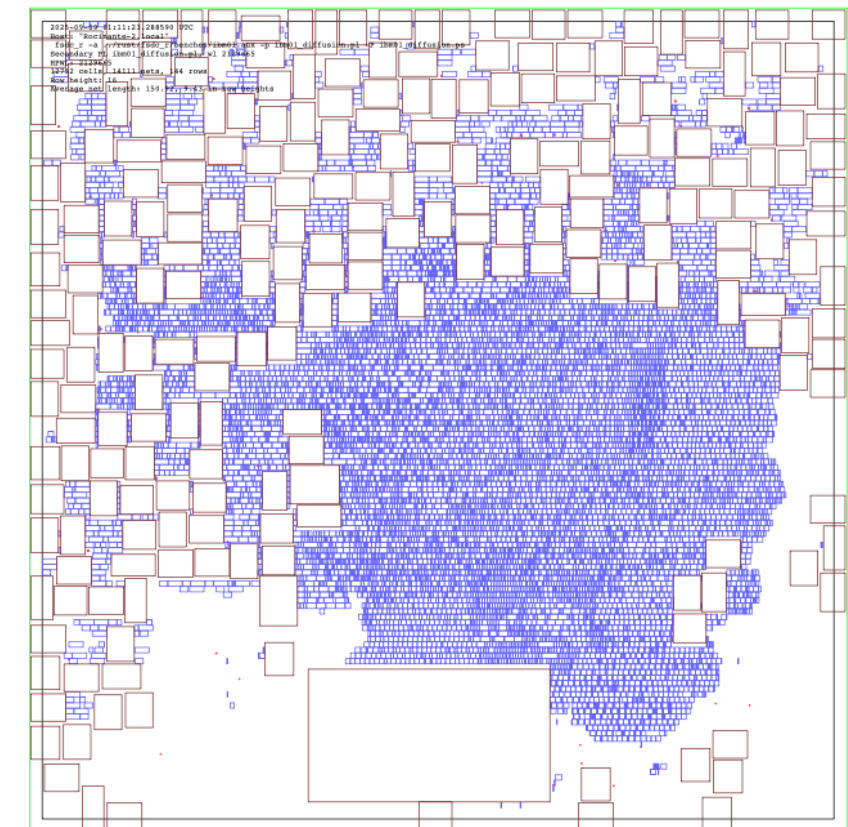
Figure 5: Placements for the ICCAD04 IBM01 benchmark, using conventional techniques. feng shui relies on recursive bisection, while NTUPlace, DREAMPlace, and RePIAce are analytic. All placements are legal, with the analytic methods obtaining the best half-perimeter wire length results.



Circuit Training: HPWL 3293660



LAMPlace: HPWL 36295401



Diffusion RL: HPWL 2129665

Figure 6: Placements for the ICCAD04 IBM01 benchmark using AI techniques. The circuit training placement was created using an independent implementation of the Google reinforcement learning approach; the placement is legal, but has significantly higher half-perimeter wire lengths compared to conventional approaches. The LAMPlace and Diffusion placements are illegal, having macro blocks overlapping, and macros out of the core placement region. Pads seem to be removed entirely from LAMPlace results, while pads are moved into the core by the Diffusion approach.

It's A Mess

- There are many papers from the AI community using physical design benchmarks.
- It's unclear which results are correct. Many are clearly wrong, and it's unclear where things are competitive.
- The differences in metrics that happened twenty-five years ago? Nothing compared to what seems to be happening now.
 - Sometimes macro blocks and cells overlap. Pads are moved (or removed). Wire lengths sometimes only consider macro blocks. Wire lengths might only consider *a subset* of macro blocks.
 - Some published results seem completely unconnected to reality.
 - Experimental results seem to have gone completely unchecked.
 - Many authors do not respond to email queries.

It's Not Just Mixed-Size Placement

I'm currently wrangling some problematic papers on floor planning. There are some TCAD submissions where I'm the AE, and the wrongness is shocking.

AI conferences are getting overwhelmed by submissions on a multitude of topics. There's no way to have reviewers who understand the submissions, so everything is getting thrown to AI for review.

There's intense FOMO in both conferences and journals. Even half-baked ideas get accepted, if they use AI.

I've tried to make Benchmarker generic, so that it can be used in other areas. We're drowning in a sea of AI slop.

What Can Be Done?

Say the quiet part out loud. If something looks wrong, say something.

We all want to have correct experimental results, so that we know what works, and what doesn't work. If something is wrong, we should step up and fix it.

Make experimental results easily visible, and in a timely fashion.
That's my goal with the Benchmark project.

But We Have So Many Cores!

With modern GPUs, we have hundreds, thousands, tens of thousands of compute cores!

Surely, all that CPU power changes everything!